THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · MEDIUM LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The HCF of 24, 36 and 60 is: 1

(a) 24 **(b)** 360 **(c)** 12 **(d)** 6

2. If α, β are zeroes of $x^2 - 7x + 10$, then $\alpha^2 + \beta^2$ is: 1

(a) 69 **(b)** 29 **(c)** 49 **(d)** 39

3. The value of x in $7x + 2y = -6$, $3x + 5y = 14$ is: 1

(a) -2 **(b)** 4 **(c)** 2 **(d)** -8

4. The sum of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) 2 **(b)** $-\frac{9}{2}$ **(c)** -2 **(d)** $\frac{9}{2}$

5. The 12th term of the AP whose 1st term is 7 and 5th term is 19 is: 1

(a) 43 **(b)** 37 **(c)** 36 **(d)** 40

6. If $DE \parallel BC$ in $\triangle ABC$, $AB = 5$ cm, $AD = 3$ cm, $AC = 10$ cm, then EC (cm) is: 1

(a) 2 **(b)** 10 **(c)** 4 **(d)** 6

7. The perimeter of the triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 8)$ is: 1

(a) 24 **(b)** 14 **(c)** 28 **(d)** 34

8. The value of $\sin^2 30^\circ + \sin^2 45^\circ$ is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{2}$ **(d)** $\frac{7}{4}$

9. A boy whose eyes are 2 m above the ground stands 18 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) 21 m **(b)** 18 m **(c)** 20 m **(d)** $20\sqrt{3}$ m

10. The length of the tangent from an external point P to a circle is 16 cm and P is 34 cm from the centre. The radius of the circle is: 1

(a) 30 cm **(b)** 50 cm **(c)** 17 cm **(d)** 18 cm

11. A square is inscribed in a circle of radius 7 cm. The area of the circle outside the square is: 1

(a) 98 cm² **(b)** 105 cm² **(c)** 154 cm² **(d)** 56 cm²

12. A tent is a cylinder of radius 3.5 m and height 4 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m²) needed to make the tent is: 1

(a) 220 m² **(b)** 137.5 m² **(c)** 225.5 m² **(d)** 88 m²

13. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 6, 9, 7, 5, 16, 7 respectively, The mean of the data is: 1

(a) 35 **(b)** 270 **(c)** 32.4 **(d)** 33.4

14. A red die and a blue die are rolled together. The probability that the sum is a multiple of 4 is: 1

(a) $\frac{5}{18}$ **(b)** $\frac{2}{9}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

15. The LCM of 6, 8 and 12 is: 1

(a) 12 **(b)** 24 **(c)** 30 **(d)** 48

16. If α, β are zeroes of $x^2 - 5x + 6$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $\frac{5}{6}$ **(b)** 30 **(c)** $\frac{6}{5}$ **(d)** $-\frac{5}{6}$

17. The value of y in $11x + 4y = 53$, $5x + 3y = 30$ is: 1

(a) 8 **(b)** 5 **(c)** -5 **(d)** 3

18. The product of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) $-\frac{9}{2}$ **(b)** 2 **(c)** -2 **(d)** $\frac{9}{2}$

19. The number of terms in the AP $5, 9, \ldots, 97$ is: 1

(a) 23 **(b)** 25 **(c)** 26 **(d)** 24

20. The areas of two similar triangles are 36 cm² and 100 cm². The ratio of their perimeters is 3: 1

(a) 5 **(b)** 8 **(c)** 25 **(d)** 15

21. The point which divides $(6, 0)$ and $(0, 10)$ in the ratio 1:1 has coordinates summing to: 1

(a) 8 **(b)** 4 **(c)** 30 **(d)** 16

22. If $\sin\theta = \frac{4}{5}$, then $\sec\theta + \tan\theta =$ 1

(a) $\frac{1}{3}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{9}{4}$ **(d)** 3

23. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 8 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $8\sqrt{3}$ m **(b)** $4\sqrt{3}$ m **(c)** $\frac{8\sqrt{3}}{3}$ m **(d)** 4 m

24. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 46 cm **(b)** 92 cm **(c)** 136 cm **(d)** 32 cm

25. The largest possible circle is cut from a square sheet of side 56 cm. The area of the sheet left over is: 1

(a) 3136 cm² **(b)** 672 cm² **(c)** 1344 cm² **(d)** 2464 cm²

26. A hemispherical bowl has outer radius 6 cm and inner radius 3 cm. The volume of material used to make the bowl is: 1

(a) 792 cm³ **(b)** 1188 cm³ **(c)** 396 cm³ **(d)** 198 cm³

27. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 6, 15, 12, 5, 2 respectively, The median of the data is: 1

(a) 26 **(b)** 30 **(c)** 20 **(d)** 25.4

28. A coin is tossed three times. The probability of getting at least one tail is: 1

(a) $\frac{7}{8}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{8}$ **(d)** 1

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29. The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is: 1

(a) 26 **(b)** 65 **(c)** 1 **(d)** 13

30. If $x = 2$ is a zero of $x^2 - 7x + k$, then k is: 1

(a) 18 **(b)** 10 **(c)** 14 **(d)** 5

31. The pair $2x + ky = 6$, $4x + 4y = 12$ has infinitely many solutions when $k =$ 1

(a) 2 **(b)** 4 **(c)** 6 **(d)** 3

32. The roots of $x^2 - 11x + 18 = 0$ differ by: 1

(a) 8 **(b)** 7 **(c)** 11 **(d)** 18

33. The sum of all two-digit multiples of 3 is: 1

(a) 1566 **(b)** 1653 **(c)** 1665 **(d)** 1683

34. The altitude to the hypotenuse of a right triangle divides it into 4 cm and 9 cm. The altitude (cm) is: 1

(a) 6 **(b)** 13 **(c)** 36 **(d)** $\frac{13}{2}$

35. The centroid of the triangle with vertices $(1, 2)$, $(4, 6)$, $(7, 1)$ has x-coordinate: 1

(a) 6 **(b)** 4 **(c)** 12 **(d)** 3

36. If $\cos\theta = \frac{3}{5}$, then $\operatorname{cosec}\theta - \cot\theta =$ 1

(a) $\frac{5}{4}$ **(b)** 2 **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

37. Two points on level ground lie on opposite sides of a tower of height $6\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) 12 m **(b)** 24 m **(c)** 48 m **(d)** $24\sqrt{3}$ m

38. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 40 cm. The length of each tangent is 9 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 360 cm² **(b)** 180 cm² **(c)** 720 cm² **(d)** 98 cm²

39. A circular park of radius 20 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 517 m² **(b)** 2068 m² **(c)** 880 m² **(d)** 1034 m²

40. A hemispherical depression of diameter 21 cm, the largest possible, is scooped out of the top face of a cubical block of edge 21 cm. The volume of the remaining block is: 1

(a) 4410 cm³ **(b)** 2425.5 cm³ **(c)** 6835.5 cm³ **(d)** 9261 cm³

41. For the data with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 2, 16, 10, 2, 4, 5, 11 respectively, The mode of the data is: 1

(a) 94.8 **(b)** 80 **(c)** 70 **(d)** 77

42. 30 cards numbered 1 to 30 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 or 5 is: 1

(a) $\frac{8}{15}$ **(b)** $\frac{13}{30}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{7}{15}$

43. The least number which when divided by 6, 8 and 12 leaves remainder 3 each time is: 1

(a) 51 **(b)** 21 **(c)** 27 **(d)** 24

44. The value of k for which $x = 2$ is a zero of $x^2 + kx - 10$ is: 1

(a) 6 **(b)** 3 **(c)** -3 **(d)** 7

45. For the pair $kx + 2y = 5$ and $6x + y = 3$ to have a unique solution, k must NOT equal: 1

(a) 10 **(b)** 8 **(c)** 12 **(d)** 3

46. If $x = 4$ is a root of $x^2 - 9x + k = 0$, then $k =$ 1

(a) 5 **(b)** -20 **(c)** 52 **(d)** 20

47. The sum of the first 14 terms of the AP $12, 17, 22, 27, \ldots$ is: 1

(a) 546 **(b)** 705 **(c)** 623 **(d)** 1246

48. A pole 6 m high casts a shadow 4 m long. A tower casting a shadow 24 m has height (m): 1

(a) 20 **(b)** 26 **(c)** 36 **(d)** 144

49. A circle has centre $(1, 2)$ and passes through $(4, 6)$. Its radius is: 1

(a) 25 **(b)** 7 **(c)** 1 **(d)** 5

50. $\dfrac{\sin 20^\circ}{\cos 70^\circ} + \dfrac{\cos 30^\circ}{\sin 60^\circ} =$ 1

(a) 0 **(b)** 2 **(c)** 4 **(d)** 1

END OF QUESTION PAPER **Total: 50 Marks**

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